12. Fractions, Decimals & Percentages

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Fraction to decimal:

Write 3/4 as a decimal.

Answer: 3/4 means 3 ÷ 4 = **0.75**.

Example 2 — Decimal to percentage:

Write 0.6 as a percentage.

Answer: 0.6 = 60 hundredths = **60%**.

Example 3 — Fraction to percentage:

Write 9/20 as a percentage.

Answer: Make the denominator 100: 9/20 = 45/100 = **45%**.

Example 4 — Percentage to fraction, simplified:

Write 35% as a fraction in its simplest form.

Answer: 35% = 35/100. Both share a common factor of 5: 35 ÷ 5 = 7, 100 ÷ 5 = 20. Simplified: **7/20**.

Example 5 — Multi-step percentage word problem:

A £60 voucher has 20% spent already. How much has been spent, and how much is left?

Answer: Find 20% of £60: (60 ÷ 100) × 20 = £12 spent. Remaining: £60 − £12 = **£48 left**.

Foundation

15 questions

1. Write 1/4 as a decimal.

2. Which fraction is equal to 0.5? A) 1/4 B) 1/2 C) 3/4 D) 1/5

3. Write 0.5 as a percentage.

4. Write 50% as a fraction.

5. What is 3/10 written as a percentage? A) 3% B) 13% C) 30% D) 33%

6. Write 1/5 as a decimal.

7. Select ALL the fractions below that are equal to 0.25: A) 1/4 B) 2/8 C) 25/100 D) 2/5 E) 1/5

8. Write 0.4 as a percentage.

9. Write 80% as a fraction.

10. What is 9/10 written as a percentage? A) 9% B) 19% C) 90% D) 99%

11. Write 7/10 as a decimal.

12. Write 0.9 as a fraction.

13. Select ALL of the following that are equal to 0.2: A) 1/5 B) 20% C) 2% D) 20/100 E) 2/10

14. Write 20% as a fraction.

15. Write 1/4 as a percentage.

Standard

20 questions

1. Order these from smallest to largest: 40%, 0.3, 1/5.

2. Write 0.05 as a fraction in its simplest form.

3. Which percentage is equivalent to 2/5? A) 20% B) 25% C) 40% D) 45%

4. Which is bigger: 30% or 0.25?

5. Select ALL of the following that are equal to 75%: A) 0.75 B) 3/4 C) 7.5 D) 75/100 E) 3/40

6. Order these from smallest to largest: 70%, 0.4, 3/10.

7. Write 0.8 as a fraction in its simplest form.

8. What is 7/10 written as a percentage? A) 7% B) 17% C) 70% D) 77%

9. Which is bigger: 35% or 0.3?

10. Write 70% as a decimal.

11. Select ALL of the values below that are LESS than 40%: A) 0.25 B) 2/5 C) 45% D) 1/5 E) 0.4

12. Write 0.6 as a fraction in its simplest form.

13. Write 3/20 as a percentage.

14. Which is bigger: 55% or 0.6? A) 55% B) 0.6 C) They are equal D) Cannot tell

15. Write 90% as a decimal.

16. Write 7/20 as a decimal AND a percentage.

17. Put these three values in order, smallest first: 45%, 0.3, 2/5.

18. A £80 voucher has 25% spent already. How much has been spent? A) £8 B) £16 C) £20 D) £25

19. 1/4 of a class of 24 children wear glasses. What percentage of the class wears glasses, and how many children is that?

20. Select ALL of the values below that are equal to 60%: A) 0.6 B) 3/5 C) 0.06 D) 6/10 E) 60/1000

Hard

5 questions

1. Write 9/20 as a decimal AND a percentage.

2. Put these three values in order, smallest first: 65%, 0.5, 3/5.

3. A £120 voucher has 15% spent already. How much money (in £) has been spent?

4. 3/10 of a class of 30 children wear glasses. What percentage of the class wears glasses, and how many children is that?

5. Select ALL of the values below that are equal to 35%: A) 0.35 B) 7/20 C) 0.035 D) 35/100 E) 3.5/10

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